RP2350 CONNECT — REV A — 85 x 70 mm, 4-layer PCB

16MB flash + 8MB PSRAM (shared QMI bus, GPIO0 CS1)

Wi-Fi / BLE: ESP32-C3-WROOM-02-N4

USB-C: USB 2.0 FS sink, independent 5.1k CC resistors

12MHz crystal + per-pin supply bypass

External 1.1V core; VREG_FB=GND, VREG_LX=NC

microSD SPI1, switched rail; enable after USB enumeration

USB_EN1/2 reset LOW: USB100; radio and SD reset OFF

Firmware: USB500 after enumeration; handle USB suspend; set REG.HIZ

LiPo 4.2V, protected pack +10k NTC; pin1=BAT+, 2=GND, 3=NTC; LiPo power-path

3.3V buck-boost; nominal 66uF output

RGB: dedicated 5V boost + AHCT level buffer

Qwiic / STEMMA QT: 3.3V I2C

Sockets expose user GPIO and power-control signals

All fitted parts assembled by PCBWay. Battery, microSD media and cables are external. See README and fab instructions.

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